An Exact Quartic Variational Period-Three Certificate

Route-A finite certificate C120

Abstract

We give an exact rational certificate for a primitive period-three orbit of an area-preserving quartic variational map. The coordinate-swap reversor and type-one generating function are verified symbolically. Chronological monodromy is a nontrivial Jordan matrix at multiplier -1, while the cyclic action has value $1/2$ and a nondegenerate index-two Hessian. Three negative controls localize the result to the frozen parameter, cubic term, and cyclic word. This is finite low-period evidence, not a global orbit, transfer, or spectral theorem.

1 Reversible variational map

Let

$$V(q) = \frac{q^4}{4} - q^2, \qquad F(q,p) = (V'(q) - p, q) = (q^3 - 2q - p, q).$$

The Jacobian, inverse, and coordinate-swap reversor R are

$$B(q) = \begin{pmatrix} 3q^2 - 2 & -1 \\ 1 & 0 \end{pmatrix}, \quad F^{-1}(Q,P) = (P, P^3 - 2P - Q), \quad R(q,p) = (p,q).$$

Direct polynomial multiplication gives $\det B(q) = 1$ and $RFR = F^{-1}$; both compositions with the displayed inverse are the identity. The type-one generating function

$$S(q,Q) = qQ - V(q)$$

recovers F under the explicit convention $p = -\partial_q S$ and $P = \partial_Q S$: namely $Q = V'(q) - p$ and $P = q$. This sign convention is part of the frozen source rather than an inferred numerical fit.

The fixed equation is $q^3 - 3q = q$, hence

$$q(q-2)(q+2) = 0,$$

and the phase-space fixed points are $(0,0), (2,2), (-2,-2)$.

2 Primitive three-cycle and tangent data

The three distinct states

$$x_0 = (0,-1), \qquad x_1 = (1,0), \qquad x_2 = (-1,1)$$

satisfy $F(x_0) = x_1$, $F(x_1) = x_2$, and $F(x_2) = x_0$. They therefore form a primitive period-three orbit. The chronological derivative is

$$M = B(-1)B(1)B(0) = \begin{pmatrix} -1 & 0 \\ -3 & -1 \end{pmatrix}.$$

Consequently

$$\operatorname{tr} M = -2, \qquad \det M = 1, \qquad \det(I - zM) = 1 + 2z + z^2 = (1+z)^2.$$

Thus the named cycle has a repeated multiplier -1 and a nontrivial Jordan part. These statements concern one finite monodromy matrix only.

3 Action and Morse certificate

Write the orbit states as (q_i, q_{i-1}) with cyclic indices and define

$$\mathcal{A}(q_0,q_1,q_2) = \sum_{i=0}^{2} \big(q_i q_{i+1} - V(q_i)\big).$$

Its stationarity equations are $q_{i-1} + q_{i+1} - V'(q_i) = 0$, exactly the orbit recurrence. At $q_* = (0, 1, -1)$, direct evaluation gives

$$\mathcal{A}(q_*) = 0 - \frac{1}{4} + \frac{3}{4} = \frac{1}{2}, \qquad \nabla\mathcal{A}(q_*) = 0,$$

and

$$H = D^2\mathcal{A}(q_*) = \begin{pmatrix} 2 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{pmatrix}.$$

The exact Hessian data are

$$\det H = 4, \qquad \det(\lambda I - H) = (\lambda + 2)(\lambda^2 - 2\lambda - 2).$$

Its eigenvalues are $-2, 1 - \sqrt{3}, 1 + \sqrt{3}$. Hence the critical point is nondegenerate and has Morse index two. The nondegenerate action Hessian and the parabolic phase-space monodromy are distinct, compatible finite certificates; neither is promoted to a global classification.

4 Controls, reproducibility, and scope

Three exact controls test attribution:

control	tested transition	actual-minus-target residual
$q^3 - \frac{5}{2}q - p$	$(1, 0) \mapsto (-1, 1)$	$(-1/2, 0)$
delete q^3	$(1, 0) \mapsto (-1, 1)$	$(-1, 0)$
word $(0, 1, 0)$	second transition	$(-1, 0)$

Each residual is nonzero, so the frozen certificate is neither parameter- agnostic nor a property of an arbitrary length-three word.

From the package directory run

```
python3 code/c120_variational_period3_producer.py
python3 code/c120_variational_period3_checker.py
python3 code/c120_sympy_crosscheck.py
python3 code/c120_replay.py
python3 code/c120_mutation.py
```

The independent checker, 29 direct symbolic checks, canonical replay, and all 21 hostile mutations pass.

The evaluator-native verdicts are

$$A1 = \texttt{A1\_WEAK}, \quad A2 = \texttt{A2\_FAIL},$$
$$A3 = \texttt{A3\_FAIL}, \quad A4 = \texttt{A4\_FORMAL\_HINT}.$$

The overall status is `ROUTE_A_EXPLORATORY`. The exact cycle and Morse data are finite structural evidence, but there is no target prime correspondence or complete enumeration, no source-owned dynamical zeta/Fredholm object, and no target divisor or global analytic structure. The generating structure is only a formal liftability hint: no quantum object, Hilbert space, or operator domain is defined. We claim no arithmetic local data, Euler factors, root numbers, automorphy, Hilbert–Pólya operator, or Route B. The firewall is `NO_BAD_EULER_OR_ROOT_NUMBER`.